

Canonical Domain Reduction for Partial Counterfactual Identification

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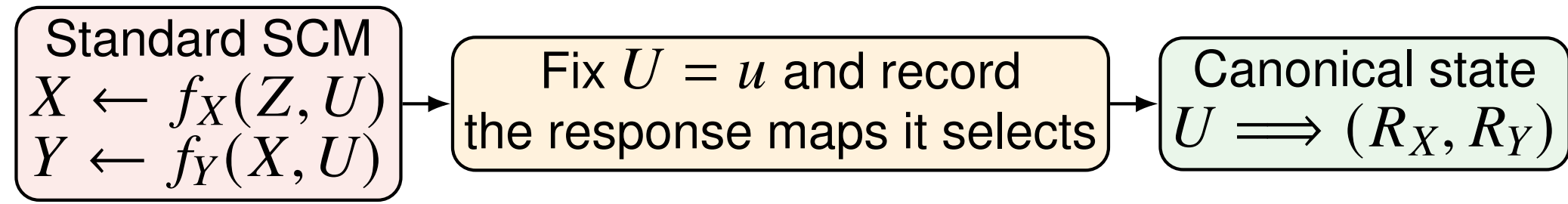
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Main message: A canonical state can be removed only if it is indistinguishable to *both* the target query and every data-implied constraint. The resulting quotient LP is smaller, yet returns **exactly the same sharp bounds**.

Canonical SCMs: A Finite Response-Type View

A general SCM $\mathcal{M} = \langle V, U, \mathcal{F}, P(U) \rangle$ may contain continuous latent variables and arbitrary structural functions, so direct optimization is infinite-dimensional.

Canonical representation. For finite observed variables, we can replace an arbitrary latent value $U = u$ by the *deterministic response functions* it selects. This gives a **finite-state canonical SCM** with exactly the same counterfactual distribution.

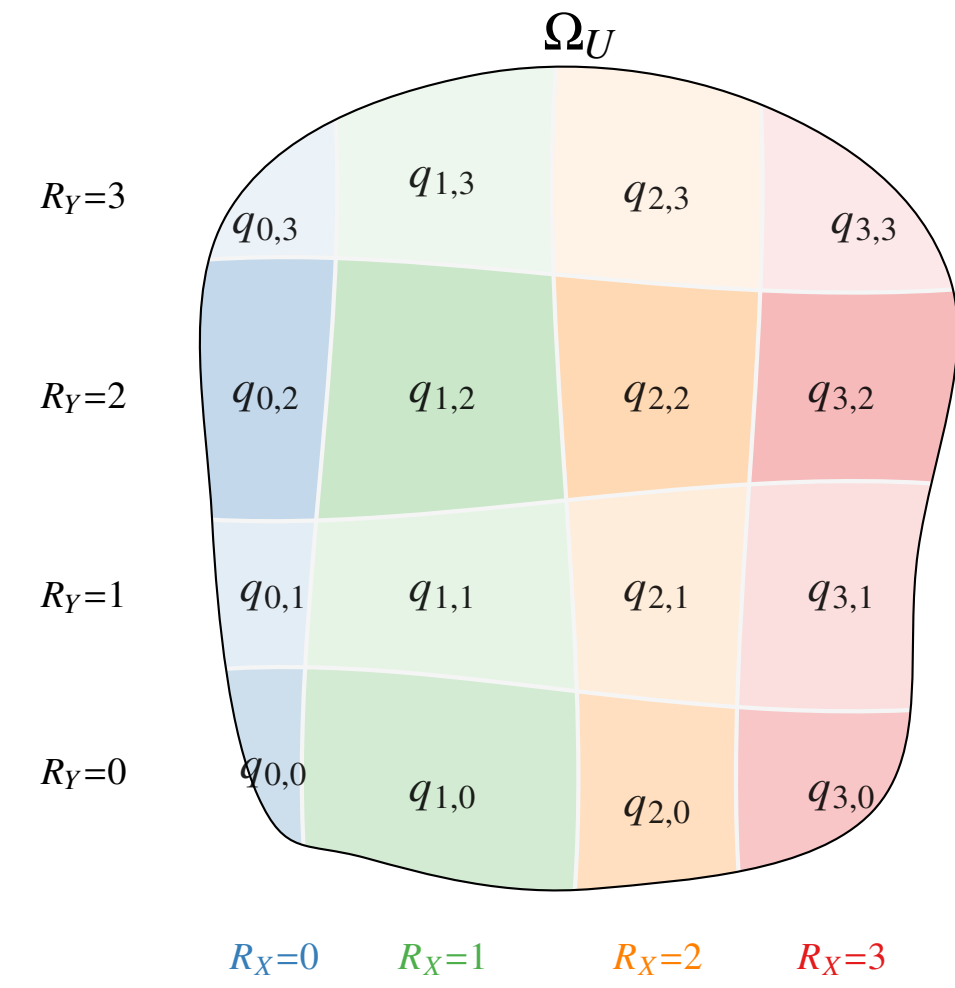


Binary response types. For binary $Z, X, Y \in \{0, 1\}$, there are four deterministic maps $Z \mapsto X$ and four deterministic maps $X \mapsto Y$.

$R_X (f_X(0, u), f_X(1, u))$	Rule	$R_Y (f_Y(0, u), f_Y(1, u))$	Rule
0	(0, 0) always 0	0	(0, 0) always 0
1	(0, 1) $X = Z$	1	(0, 1) $Y = X$
2	(1, 0) $X = 1 - Z$	2	(1, 0) $Y = 1 - X$
3	(1, 1) always 1	3	(1, 1) always 1

$$q_{j,k} = P(R_X = j, R_Y = k), \quad \sum_{j=0}^3 \sum_{k=0}^3 q_{j,k} = 1.$$

Joint canonical domain.



A **canonical state** is one cell (R_X, R_Y) in this finite domain, and its probability mass $q_{j,k}$ is the optimization variable.

Baseline Sizes Across $\mathcal{L}_1, \mathcal{L}_2, \mathcal{L}_3$

For U indexing c-component $C(U)$, higher PCH layers encode responses to more parent configurations:

$\mathcal{L}_1$: observational

$\mathcal{L}_2$: single-world

$\mathcal{L}_3$: counterfactual

$$d_U^{(1)} = \prod_{V \in \text{Pa}(C(U))} |\Omega_V|, \quad d_U^{(2)} = \prod_{V \in C(U)} |\Omega_{\text{Pa}_V} \times \Omega_V|, \quad d_U^{(3)} = \prod_{V \in C(U)} |\Omega_{\text{Pa}_V} \mapsto \Omega_V|$$

where $|\Omega_A \mapsto \Omega_B| = |\Omega_B|^{|\Omega_A|}$. For the binary NDE graph, $C_1 = \{X, W\}$ and $C_2 = \{Y\}$:

$$|\Omega^{\mathcal{L}_1}| = 4 \times 8 = 32, \quad |\Omega^{\mathcal{L}_2}| = 8 \times 8 = 64, \quad |\Omega^{\mathcal{L}_3}| = 8 \times 16 = 128.$$

These are layer-wise sufficient domains; the reductions below additionally depend on the target query and information set.

Sharp Bounds and the Safe Quotient

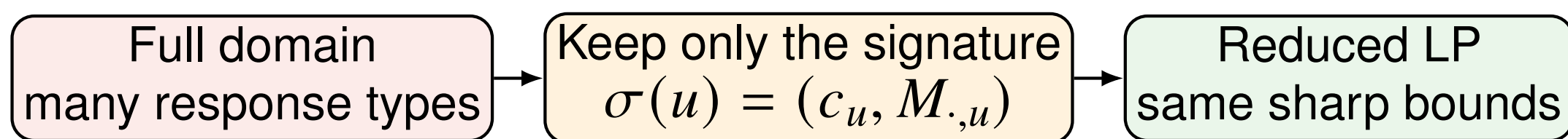
For each full state u , the query and every data-implied quantity are fixed. Hence

$$\underline{Q}, \bar{Q} = \min_{\psi \geq 0} / \max_{\psi \geq 0} c^\top \psi \quad \text{s.t.} \quad M\psi = p, \quad 1^\top \psi = 1.$$

The LP sees state u only through

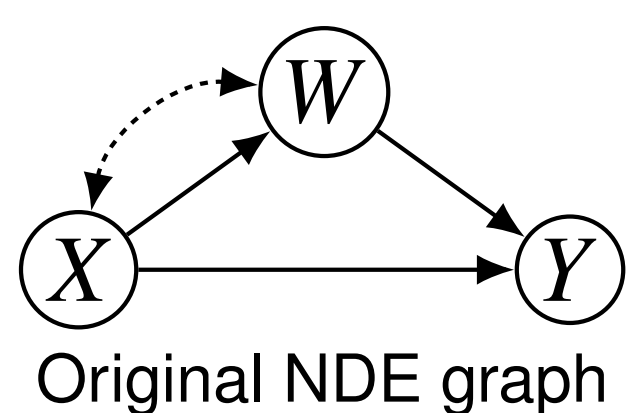
$$\sigma(u) := (c_u, M_{\cdot, u}), \quad u \sim u' \iff \sigma(u) = \sigma(u'), \quad \phi_t = \sum_{u: \pi(u)=t} \psi_u.$$

Merging equal-signature states therefore preserves feasibility and both sharp endpoints exactly.

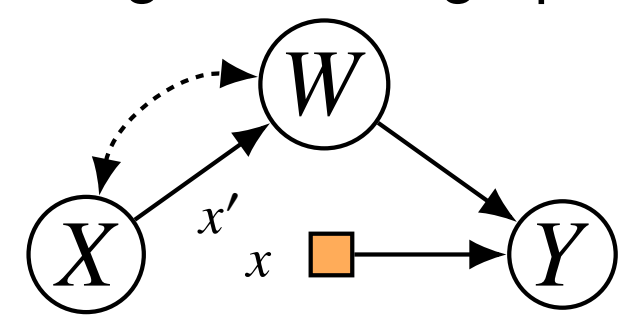


Why query-only merging is unsafe. The same query value c_u may accompany different observed-data values $M_{\cdot, u}$, so ignoring constraints can change the bounds.

Running Example: NDE Query



Original NDE graph



Realizable graph for $Y_{x, w_{x'}}$

For the binary mixed-world query

$$Q = P(Y_{0, w_1} = 1),$$

set $A = W_1$, $B = Y_{0,0}$, and $C = Y_{0,1}$. Then

$$Y_{0, w_1} = (1 - A)B + AC, \quad Q = g^\top \phi,$$

so the query itself needs only the joint distribution of $(W_1, Y_{0,0}, Y_{0,1})$.

Information regime	Response entries retained	Size
Worst-case $\mathcal{L}_3$	$X, W_0, W_1, Y_{00}, Y_{01}, Y_{10}, Y_{11}$	128
Full P_{obs}	$X, W_0, W_1, Y_{00}, Y_{01}, Y_{1, w_1}$	64
$\mathcal{L}_{2,5}$ marginals	W_1, Y_{00}, Y_{01}	8

Full P_{obs} : why 16 signature classes? Only the observed cell (X, W_X, Y_{X, w_X}) and query value Y_{0, w_1} matter, giving $8 \times 2 = 16$.

Key point: the effective canonical size is determined jointly by the target query Q and information set $\mathcal{I}$.

Query-Specific Context Counting

We can often predict the reduced size before constructing the full domain. Let $\mathcal{T}_V(Q)$ collect the labeled parent assignments at which f_V is evaluated, and define

$$S_V(Q) := \left\{ x = ((p = a_p)_{\lambda_p})_{p \in \text{pa}(V)} : x \in \mathcal{T}_V(Q), \lambda_p = \lambda_Q(p) \forall p \in \text{pa}(V) \right\}, \quad m_V(Q) := |S_V(Q)|.$$

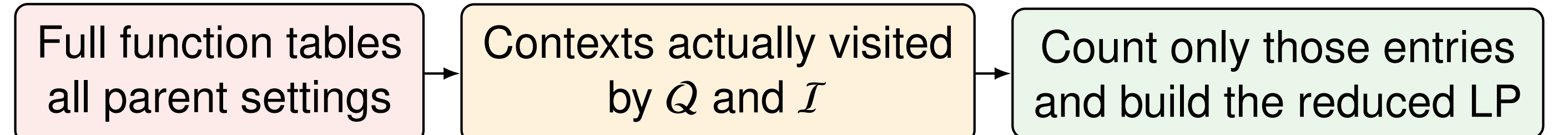
A sufficient function-space cardinality is

$$d_U = \prod_{V \in C(U)} |\Omega_V|^{m_V(Q)}.$$

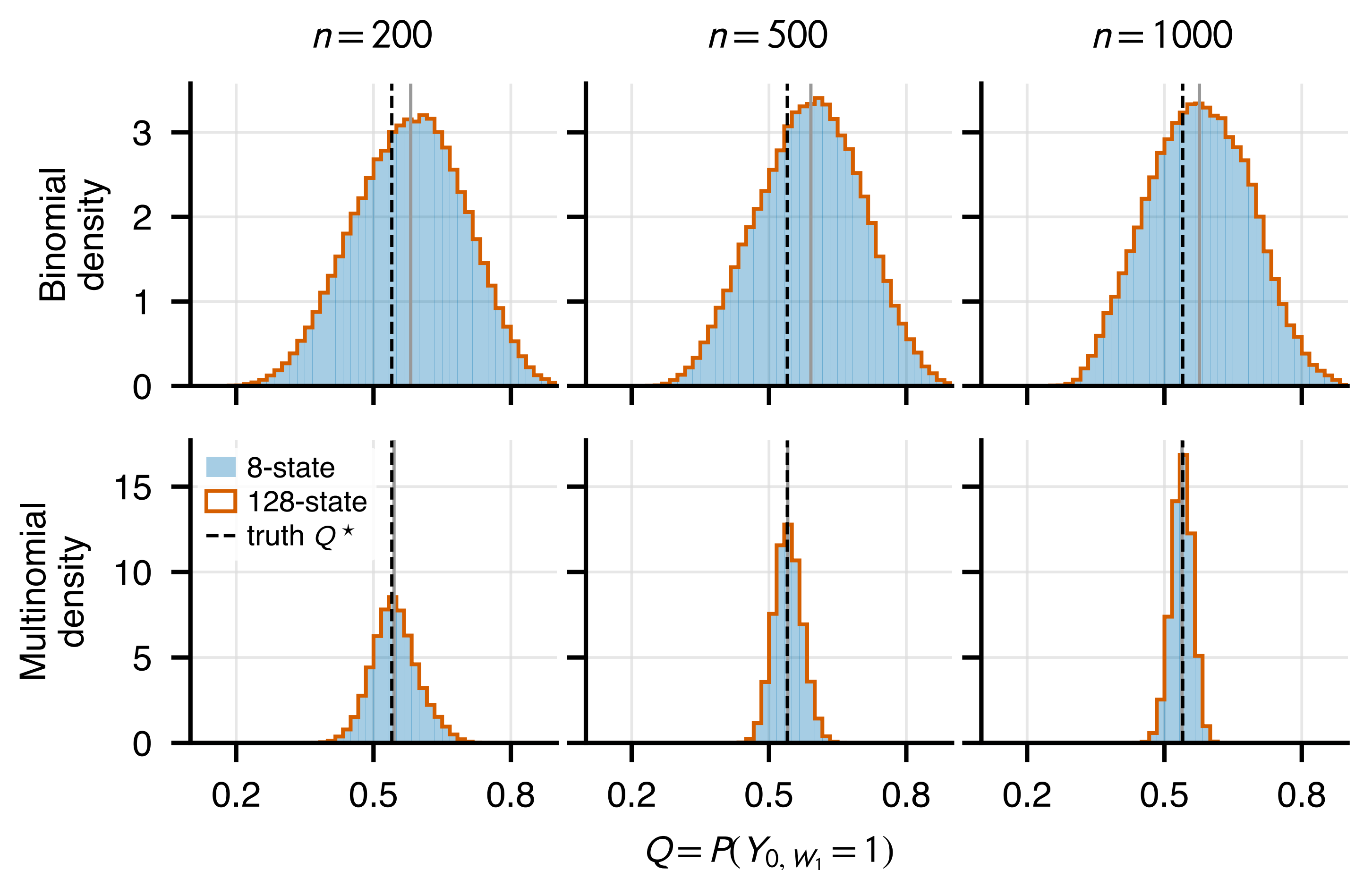
For $Q = P(Y_{x, w_{x'}})$,

$$S_X(Q) = \emptyset, \\ S_W(Q) = \{(x')\}, \\ S_Y(Q) = \{(x, 0_{x'}), (x, 1_{x'})\}.$$

Hence $m_X = 0$, $m_W = 1$, and $m_Y = 2$, giving $2^1 \times 2^2 = 8$ states instead of constructing all 128 states.



Posterior Propagation: 8 vs. 128 States



Posterior histograms under marginal-binomial data (top) and direct query-state multinomial data (bottom). The reduced 8-state and lifted 128-state evaluations overlap up to numerical precision.

9.42×
solver-time speedup
128 → 8

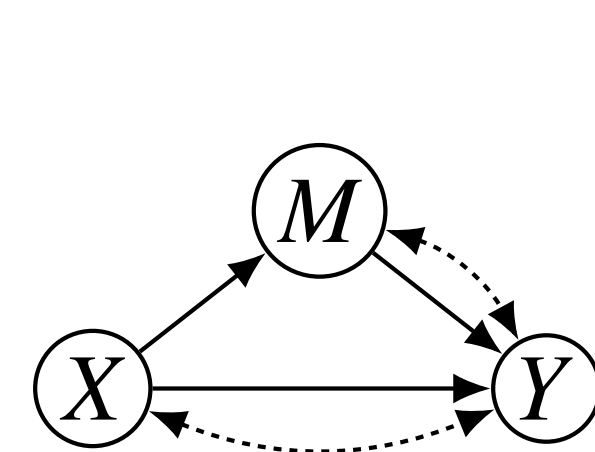
300.6×
stress-test speedup
2048 → 8

$\sim 10^{-16}$
query-lift discrepancy

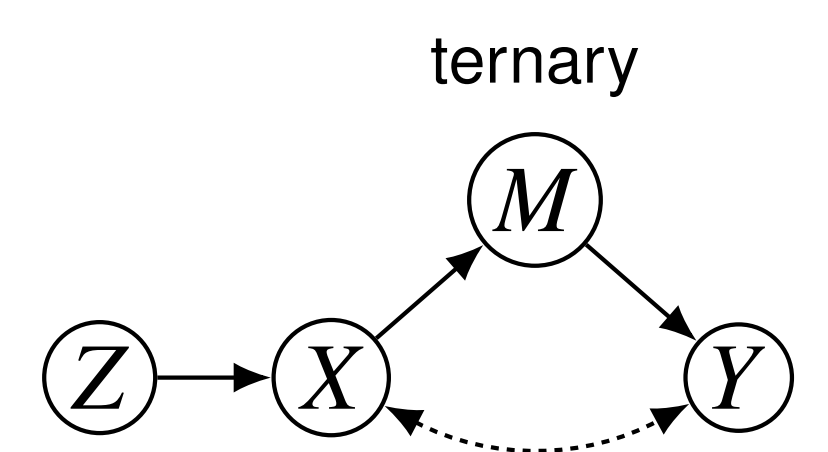
	Reduced	Lifted	Speedup
NDE sharp-bound LP	8 states, 0.036 ms	128 states, 0.341 ms	9.42×
Stress scaling	8 states, 0.035 ms	2048 states, 10.43 ms	300.6×

The 8-state formulation is not an approximation.
For this query–information pair, it is the exact quotient of the 128-state formulation.

Larger Graphs



PNS confounded mediator



NDE-type ternary mediator

Graph	Lifted	Reduced	Compression	Speedup	Max gap
PNS graph	128	12	10.7×	6.89×	2.3×10^{-9}
Ternary graph	4608	48	96×	118.6×	3.4×10^{-9}

The same idea scales beyond the running example: safe quotients keep the sharp bounds while dramatically shrinking larger lifted domains.

Takeaways

Finite: canonical SCMs turn partial identification into optimization over deterministic response types.

Exact and query-aware: safe quotients preserve feasibility and both endpoints, while context counting adapts the domain to the query and information set.

Practical: much smaller LPs return the same answers.